

Use-Case Flow Specifications (Multi-Use Case Package)

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #07 — Campus & Academic Operations

UC-04: Record Closure Sign-off (FR-004 / Close Ticket)

Primary Actor: Hostel Resident	Supporting Actors: Maintenance Warden / Staff
Preconditions: Ticket status set to 'Resolved'.	Postconditions (Success): Resident sign-off recorded, status set to 'Closed'.

Main Success Scenario

- System sends notification to resident once ticket status is marked 'Resolved'.
- Resident inspects physical repair work completed in room.
- Resident accesses ticket on app dashboard and selects 'Confirm Resolution & Sign-Off'.
- Resident provides digital signature/confirmation and optional 1-5 star satisfaction rating.
- System validates sign-off input, links resident ID + timestamp to ticket record, and updates status to 'Closed' (include relationship execution).
- System archives ticket record into historical logs. Use case ends.

Alternate Flow — 3a: Resolution Rejected

- 3a. Triggered at step 3 if resident finds repair incomplete or unsatisfactory.
- 3a1. Resident selects 'Reject Resolution' and enters detailed feedback.
- 3a2. System reverts ticket status back to 'In Progress', notifies Warden queue, and increments escalation priority. Use case ends.

Requirements Table

	A	B	C	D	E	F
1	Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
2	FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity (Plumbing/Electrical/Carpentry; Low/Medium/High).	High	Pass: Ticket is created, assigned a unique ID, and routed to the warden queue. Fail: Incomplete form (e.g., missing room tag) is accepted.	Core intake function; accurate tagging and severity are required to route and prioritize the ticket.
3	FR-002	Functional	The system shall allow the Maintenance Warden to assign a submitted ticket to a specific maintenance staff member and set a target resolution SLA based on issue severity.	High	Pass: Warden selects a staff member from a list, ticket status changes to "Assigned," and the assigned staff is notified within 1 minute. Fail: Ticket remains unassigned after the warden submits the assignment.	Ensures every logged issue is routed to a responsible person for timely resolution.
4	FR-003	Functional	The system shall allow the Maintenance Warden or assigned staff to update a ticket's status (Open → In Progress → Resolved) as work progresses.	High	Pass: Each status change is timestamped and visible on the resident's ticket view within 1 minute. Fail: A status change is not reflected to the resident.	Keeps residents informed and creates an auditable timeline needed for SLA tracking.
5	FR-004	Functional	The system shall require the reporting resident to confirm and record a closure sign-off before a ticket can be marked "Closed."	Medium	Pass: A ticket cannot transition to Closed without a stored sign-off (resident ID + timestamp). Fail: A ticket is marked Closed with no sign-off record.	Verifies the issue is actually resolved from the resident's perspective and creates accountability for maintenance staff.
6	FR-005	Functional	The system shall allow residents to view the real-time status and history of all maintenance tickets they have submitted.	Medium	Pass: The resident's dashboard lists every submitted ticket with current status and last-updated timestamp. Fail: A submitted ticket is missing from the resident's list.	Transparency reduces duplicate ticket submissions and manual status-check requests to the hostel office.
7	NFR-001	Performance & Security	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without a status change.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load, and escalation alerts fire within the 24-hour window.	Prevents high-severity issues (e.g., electrical hazards) from being silently ignored past the SLA.
8	NFR-002	Nonfunctional — Usability	The system shall allow a resident to complete and submit a new ticket, including a photo attachment, within 90 seconds on a standard smartphone browser.	Medium	Pass: Usability testing with 10 residents shows average submission time ≤ 90s. Fail: average exceeds 90s.	Low reporting friction increases adoption and reduces issues going unreported.

UseCase Diagram

UML Use-Case Diagram: Hostel Maintenance & Issue Ticketing System

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